

Q-01: With the advent of quantum computing, traditional public key cryptosystems such as RSA and ECC are potentially vulnerable.

Answer: Quantum computing threatens traditional public-key cryptosystems like RSA and ECC because Shor's algorithm can efficiently factor large integers and solve discrete logarithms, breaking security. To address this, post-quantum cryptography (PQC) proposes alternatives such as lattice-based.

Q-02: Design and implement a novel Pseudo Random number generator (PRNG) algorithm.

Answer: Below is a simple PRNG design and implementation in Python using the current timestamp, process ID, and a modulus operator.

```
python.py
import time
import os

def simple_prng(modulus=1000):
    seed = int(time.time() * 1000) ^ os.getpid()
    random_number = (seed * 1103515245 + 12345) % modulus
    return random_number.
```

Q-03 : Compare traditional ciphers with modern symmetric ciphers like AES and DES.

Answer : Traditional ciphers (Caesar, Vigenere, etc.) use simple substitution or polyalphabetic techniques with some key, making them easy to understand. Modern symmetric ciphers like DES and AES use complex mathematical operations, longer key lengths.

Q-04 : Let S_4 be the symmetric group on the set $(1, 2, 3, 4)$. Define an action of S_4 on the set of 2-element subsets of $(1, 2, 3, 4)$.

Answer : Define the action of S_4 on the set of 2-element subsets $\{1, 2, 3, 4\}$ by

$$g \cdot \{a, b\} = \{g(a), g(b)\}, \quad g \in S_4$$

The action is well-defined because permutations are bijections.

The orbit of $\{1, 2\}$ consists of all 2-element subsets of $\{1, 2, 3, 4\}$ so, $|\text{orbit}(\{1, 2\})|$
 $= \binom{4}{2} = 6$

Since $|S_4| = 24$, by the orbit-stabilizer theorem

$$|\text{stab}(\{1, 2\})| = \frac{24}{6} = 4$$

Q-05: Let $GF(2^2)$ be the finite field order 4 constructed using the irreducible polynomial $x^2 + x + 1$ over $GF(2)$.

Answer: Let $GF(2^2) = \{0, 1, \alpha, \alpha + 1\}$
where $\alpha^2 + \alpha + 1 = 0$ (so $\alpha^2 = \alpha + 1$)

1. Group under multiplication:

The nonzero elements $\{1, \alpha, \alpha + 1\}$ are closed under multiplication, multiplication is associative, 1 acts as the identity.

2) cyclic:

The multiplication group has order 3. Since $\alpha^3 = 1$, the set $\{1, \alpha, \alpha + 1\}$ is generated by α , so it is cyclic.

Q-06: Let $GL(2, R)$ be the general linear group of 2×2 invertible matrices over R . Show that the set of scalar matrices forms a normal.

Answer: Let $Z = \{ \lambda I \mid \lambda \in R^* \}$
be the set of nonzero scalar matrices in $GL(2, R)$
normal subgroup:

Z is a subgroup since it's closed under multiplication and inverses. for any $A \in GL(2, R)$ and $\lambda I \in Z$

$$A(\lambda I)A^{-1} = \lambda(AA^{-1}) = \lambda I \in Z,$$

so Z is invariant under conjugation.

Hence, $Z \triangleleft GL(2, R)$

Q-07: Explain the Diffie-Hellman key exchange protocol and its application is secure communication. Discuss the security of the Diffie-Hellman common attacks.

Answer: Diffie-Hellman key exchange allows two parties to establish a shared secret over an insecure channel. They publicly agree on a large prime p and generator g , exchanging values $g^a \bmod p$ and $g^b \bmod p$ and both compute the same secret key $g^{ab} \bmod p$. Its security relies on the discrete logarithm problem.

Q-08: Let G be a group and let H be a subgroup of G . Prove that the intersection of any

Answer: Let H_1 and H_2 be subgroups of G .

We show $H_1 \cap H_2$ is a subgroup, use the subgroup

Identity: Since both H_1 and H_2 are subgroups, $e \in H_1$ and $e \in H_2$ so $e \in H_1 \cap H_2$.

Closure: If $x, y \in H_1 \cap H_2$ then $x, y \in H_1$ and $x, y \in H_2$.
Thus, $H_1 \cap H_2$ is a subgroup of G .

Q-09: Prove that the ring $\mathbb{Z}_n$ is commutative and identify whether it has zero divisors.

Answer: Let, $\mathbb{Z}_n = \{0, 1, 2, \dots, n-1\}$ with addition and multiplication modulo n .

Commutativity: Addition and multiplication in $\mathbb{Z}_n$ are defined modulo n , inherited from integers, since integer addition and multiplication are commutative, for all $a, b \in \mathbb{Z}_n$,

$$a + b \equiv b + a \pmod{n}, \quad ab \equiv ba \pmod{n}$$

Hence, $\mathbb{Z}_n$ is a commutative ring.

Q-10: Explain the vulnerabilities of the DES chipkey and why it is considered insecure for modern use. Discuss the role of brute force attacks.

Answer: Vulnerabilities of DES: DES is considered insecure mainly due to its short key length (56 bits) which makes it vulnerable to brute force attacks can try all possible keys with modern computing power in a feasible time. DES also has known weaknesses against linear, differential and differential cryptanalysis further reducing its security margin. A shorter key drastically lowers the number of possible keys.

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Q-11: Differential cryptanalysis is a widely known attack against block cipher.

i. Explain how the Feistel structure of DES

Answer: DES and Feistel structure
DES use a Feistel structure, where data is split into two halves and processed through multiple rounds. Each round introduces nonlinearity via S-boxes and mixes data between halves which helps diffuse input differences across rounds. Although DES was designed to offer some resistance to differential.

Q-12: Use the Extended Euclidean Algorithm demonstrate how to find the modular inverse of an integer a , modulo n . How is this algorithm

Ans: For coprime integers a and n the extended Euclidean algorithm finds integers x, y such that $ax + ny = \gcd(a, n) = 1$

Reducing modulo n gives $ax \equiv 1 \pmod{n}$
so x is the modular inverse of a mod n .

In RSA key generation, the private key d is computed as the modular inverse of the public exponent e modulo $\phi(n)$
i.e., $ed \equiv 1 \pmod{\phi(n)}$.

Q-13: Consider the following modes of operation for block cipher. ECB, CBC and CTR.

Answer: Let $E_k(\cdot)$ be a block cipher cipher with block size n .

In ECB mode, encryption is defined as:

$$C_i = E_k(P_i)$$

for each plaintext block P_i

If the plaintext contains redundancy, then there exist blocks $P_i = P_j$ for $i \neq j$

$$C_i = E_k(P_i) = E_k(P_j) = C_j$$

Q-14: Why the linearity of LFSRs makes them vulnerable to known-plaintext attack.

Answer: An LFSR (Linear feedback shift Register) generates keystream bits using linear recurrence relations over $GF(2)$.

Because of this linearity, small amount of known plaintext reveals corresponding keystream, which leads to a system of linear equations. Allow to recover the LFSR's feedback polynomial and initial state efficiently.

Q-15: Let M be the set of all possible plaintexts, K the set of keys, and C the set ciphertext in a cryptographic system.

Answer: A cryptosystem has perfect secrecy if the ciphertext given no information about the plaintext. $\forall m \in M$ and $c \in C$

$$P(M=m | C=c) = P(M=m)$$

① One-time pad: $C = M \oplus K$

In the one-time pad: $C = M \oplus K$,

Q-16: A Linear Congruential generator (LCG) is defined by the recurrence relation: $x_{n+1} = (ax_n + c) \bmod m$

Answer: Choose specific parameters for

$$a=5, c=3, m=16, x_0=7$$

The LCG is defined by

$$x_{n+1} = (ax_n + c) \bmod m$$

Now compute

$$x_1 = (5 \cdot 7 + 3) \bmod 16 = 38 \bmod 16 = 6$$

$$x_2 = (5 \cdot 6 + 3) \bmod 16 = 33 \bmod 16 = 1$$

$$x_3 = (5 \cdot 1 + 3) \bmod 16 = 8 \bmod 16 = 8$$

$$x_4 = (5 \cdot 8 + 3) \bmod 16 = 43 \bmod 16 = 11$$

Q-17 : Define a ring in abstract algebra and explain its key properties. Provide an example of a commutative ring.

Answer : A ring $(R, +, \cdot)$ is a set equipped with two operations such that.

* $(R, +)$ is an abelian group

* multiplication is associative.

finite fields are constructed as quotient rings of polynomial ring e.g.

$\mathbb{F}_p[x] / (f(x))$ when $f(x)$ is irreducible

Q-18 : Given an RSA key pair with the public key (e, n) and the private key d , where: $p = 5$, $q = 11$ (two prime numbers)

Answer : RSA encryption and decryption

Given $p = 5$, $q = 11$

$$n = pq = 55 \quad \phi(n) = (5-1)(11-1) = 40$$

Choose a public exponent $e = 3$

Such that $\gcd(3, 40) = 1$

Encryption $ed \equiv 1 \pmod{40} \Rightarrow d = 27$

Message - $M = 2$

Decryption $C = M^e \pmod{n} = 2^3 \pmod{55} = 8$

$$M = C^d \pmod{n} = 8^{27} \pmod{55} = 2$$

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Q-19 : Given the elliptic curve equation
 $y^2 = x^3 + ax + b \pmod{p}$ where $p=23, a=1, b=1$

Answer : Given the elliptic curve

$$y^2 \equiv x^3 + x + 1 \pmod{23}$$

verification of point $P = (3, 10)$

$$\text{LHS : } y^2 = 10^2 = 100 \equiv 8 \pmod{23}$$

$$\text{RHS : } 3^3 + 3 + 1 = 27 + 3 + 1 = 31 \equiv 8 \pmod{23}$$

Q-20 : Suppose an elliptic curve
 $y^2 = x^3 + 7x + 10 \pmod{37}$ is used for
 ECDSA, with the base point $G = (2, 5)$ and
 order $n = 19$.

Answer : We work over the elliptic
 curve $E : y^2 \equiv x^3 + 7x + 10 \pmod{37}$
 with base point $G = (2, 5)$ of order
 $n = 19$

public key generation

• private key $d = 9$

compute public key $Q = dG = 9G$

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Q-21: Explain the key properties of cryptographic hash functions such as those in the secure hash algorithm.

Answer: Essential characteristics of secure
Pre-image resistance: Given a hash h , it is computationally infeasible to find a message m such that $H(m) = h$.

Impact of hash output length (SHA-256)

2^{256} security against pre-image attack.
 2^{128} security against collision attacks

Q-22: Explain the concept of Galois field focusing on $GF(p)$ and $GF(2^n)$. How are Galois fields used in the construction of cryptographic.

Answer: Galois fields (finite fields) are algebraic structures with a finite number of elements in which addition, multiplication, and inversion are defined. $GF(p)$: A field with p elements.
 $GF(2^n)$: A field with 2^n elements; construction using polynomials over $GF(2)$ modulo an irreducible polynomial.

Elliptic curve cryptography (ECC) Elliptic curves are defined over $GF(p)$ or $GF(2^n)$ and necessary operations rely on efficient and well-defined arithmetic.

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Q-23: Lattice-based cryptography is considered a promising candidate for post-quantum cryptography due to its.....

Answer: Shortest Vector Problem (SVP)
The shortest vector problem (SVP) asks for the shortest non-zero vector in a high-dimensional lattice. It is computationally hard and forms the core security assumption of lattice-based cryptography. Breaking schemes like lattice RSA and ECC rely on integer factorization and the discrete logarithm problem.

Q-24: In a stream cipher, let the keystream $k = (k-1, k-2, k-3, \dots)$ be generated using a linear feedback shift register (LFSR) defined by the rec..

Answer: Let the LFSR have length n and characteristic polynomial

$$f(x) = x^n + c_1 x^{n-1} + \dots + c_n$$

over $\text{GF}(2)$: If $f(x)$ is primitive then one of its roots α is a primitive of the field $\text{GF}(2^n)$.

Q-25: In lattice-based cryptography, particularly using the Learning with Error (LWE) problem and shortest vector problem (SVP) sign a message as follows:
Explain the process of signing a message as

Answer: LWE-based digital signature process:

- * choose a secret vector s (private key) with small entries.
- * Generate a random matrix A and an error vector e .
- * Compute the public key $P_k = (A, b = As + e \text{ mod } q)$
- Hash the message M to a short vector $h = H(M)$
- * use the secret key s to generate a signature vector α the satisfier a relation involving

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